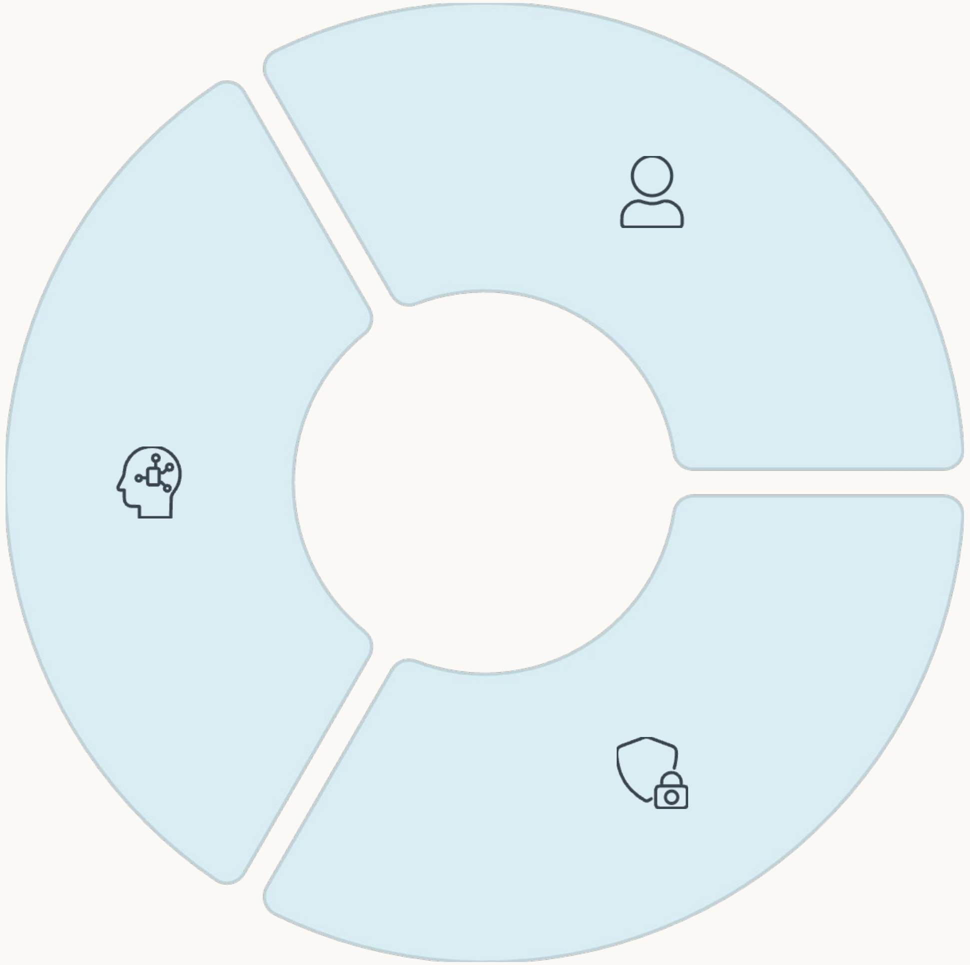
ScubaMind – Early Mental Wellbeing Through Passive Insights

ScubaMind is an AI-powered mobile application that enables early detection of depression and mental wellbeing risks through passive behavioral data and minimal user interaction.

Our Innovative Approach



User Value

Sustainable mental health tracking with early detection and minimal effort

Technology

Smart analysis powered by AI, LLM, and mobile sensor technologies

Ethics & Privacy

Your data is always protected with a secure and privacy-focused design

Critical Problems We Solve

Late Awareness

Mental state changes often go unnoticed until they reach crisis levels

High Effort

Current applications require continuous manual data entry and intense user participation

Low Sustainability

There are serious shortcomings in long-term use and effective personalization

Reactive Approach

Psychological support systems intervene only after a problem arises, not preventatively

Our Solution: Smart and Passive Tracking

How It Works?

1 **Passive Data Collection**

Sleep, step count, and daily routines are automatically monitored

2 **Low-Effort Interaction**

Voice check-ins and simple goal tracking with minimal user burden

3 **AI Analysis**

LLM-powered artificial intelligence analyzes behavioral patterns

4 **Personalized Insight**

Customized feedback and early warning systems

Our Value Proposition

Sustainable use: Natural integration into daily life

Early awareness: Detecting problems before they become critical

Personalized support: Tailored recommendations for you

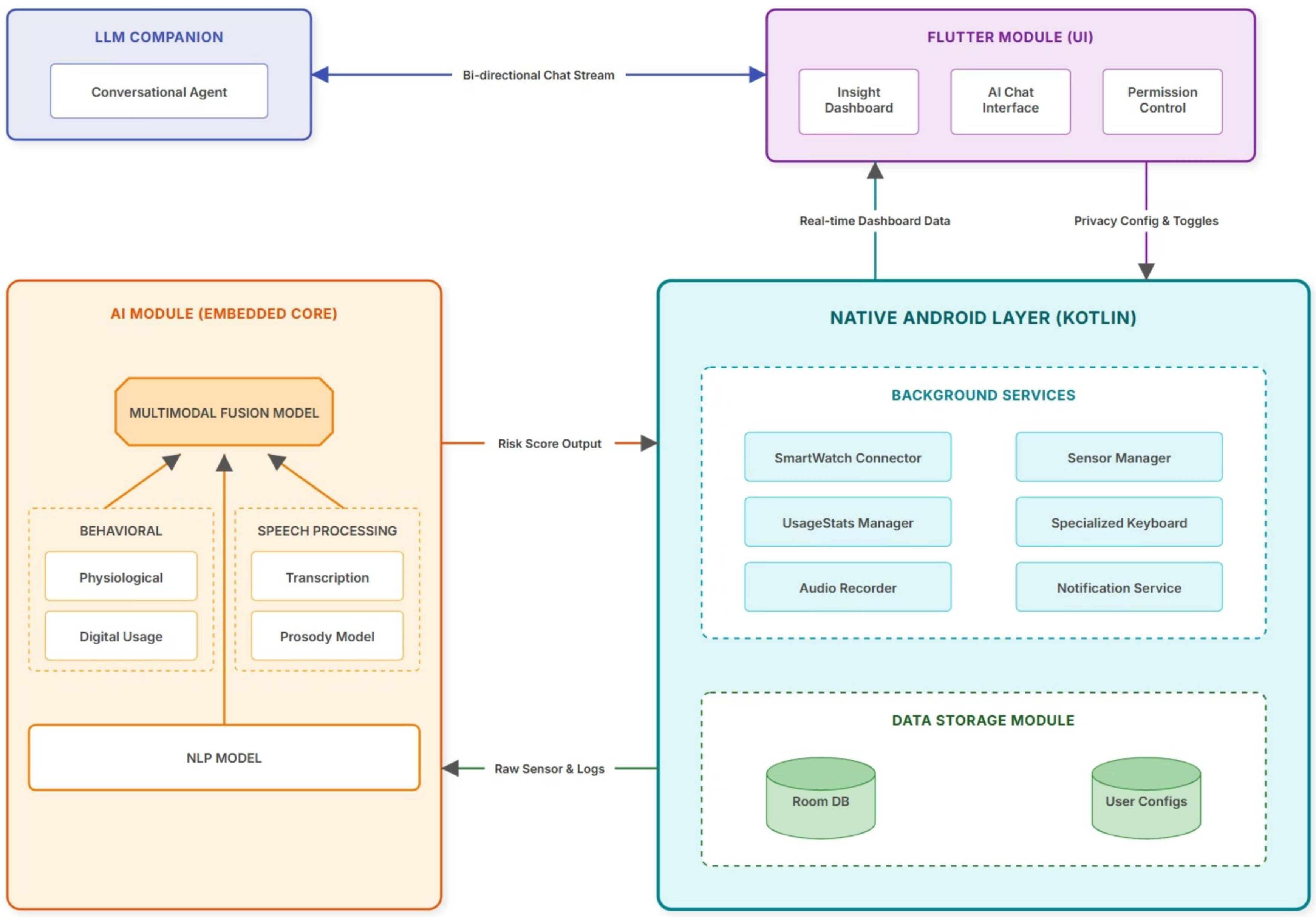
What Makes ScubaMind Different?

Passive-first design: No constant questionnaires or intrusive interactions

Early risk detection: Identifying mental health risks before symptoms become severe

Privacy-aware architecture: User data is securely handled and always protected

Data Processing Pipeline



Project Timeline

1 — September–October

Research and requirements analysis: Literature review, psychology clinic consultations, and problem definition

2 — November

System architecture and design process: High-level architecture, data flow, and privacy-focused design

3 — December–February

App development phase: Implementation of core system components (mobile app, sensor services, AI module, and database) was initiated and ongoing

4 — March–April

Integration and comprehensive testing process: Integration of core system components, system-level testing, and iterative refinements based on early development results

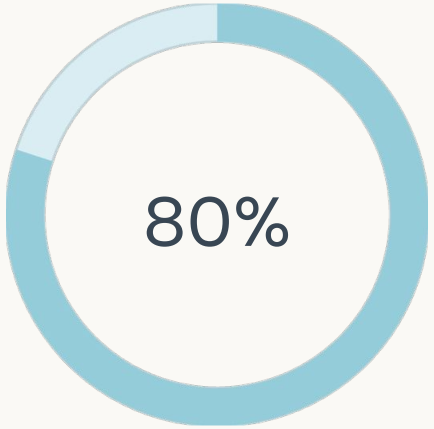
5 — May

Finalization and evaluation: System stabilization, evaluation of implemented features, documentation, and final project delivery

Current Status: The research and design phases have been successfully completed.
Core system implementation has been initiated, and we are currently in the development and integration phase

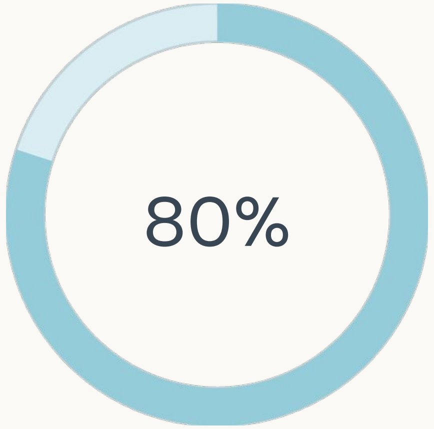
Project Status

Development Phases



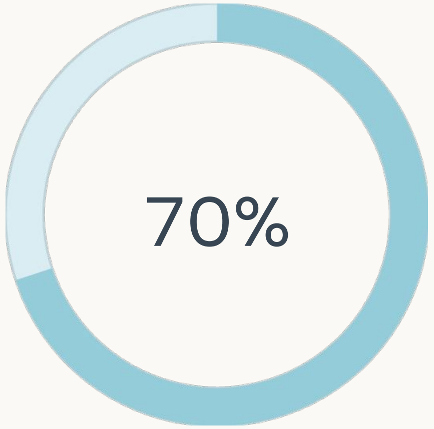
Research

In Progress



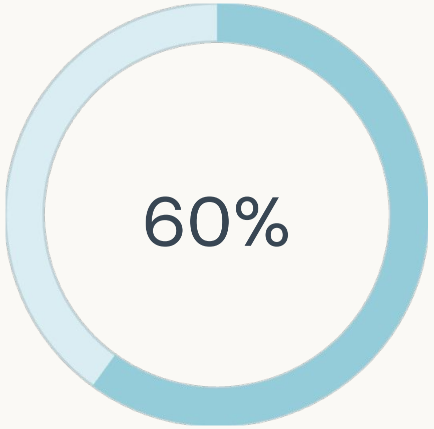
Design

In Progress



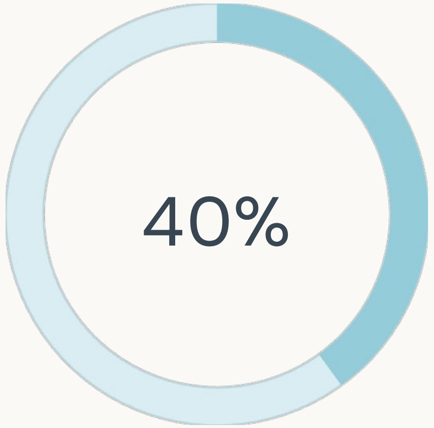
Mobile App (Flutter)

In Progress



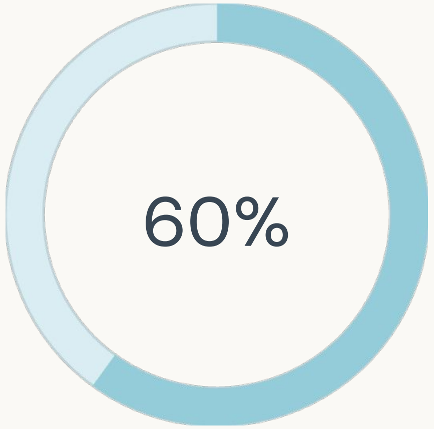
Native Android Services

In Progress



AI Module

In Progress



Overall Project Completion

In Progress

Project Health

Overall Project Health

Status: On Track. **Overall Completion:** ~60%. **Current Phase:** Development & Integration

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Key Risks & Mitigation

Risk 1 – Data Quality Passive behavioral data may include noise and inconsistencies
→ Mitigation: Iterative testing and sensor-level validation

Risk 2 – AI Model Calibration Limited labeled data for early-stage depression signals
→ Mitigation: Psychology clinic consultations and expert feedback

Risk 3 – Integration Complexity Integration of Flutter UI, native Android services, and AI modules
→ Mitigation: Modular architecture and staged integration

Ethics & Privacy Status

- Privacy-by-design approach applied
- On-device processing prioritized
- No critical ethical risks identified

Team and Role Distribution

Mobile Application Development (Flutter and Kotlin)

User interface design, onboarding flow, goal tracking, and visualization of behavioral insights within the Flutter mobile application.

Responsible: Merve Güleç, Yiğit Koşum

Native Android & Sensor Services

Implementation of native Android foreground services for passive data collection (sleep, steps, activity), sensor management, and local data handling.

Responsible: Yiğit Koşum, Murathan Işık

AI & Behavioral Analysis

Design and implementation of AI and LLM-based analysis pipelines for extracting behavioral patterns and generating early mental wellbeing insights.

Responsible: Muhammed Fatih Başal, Metin Çalışkan

System Architecture & Privacy Design

Overall system architecture, data flow design, privacy-by-design principles, and secure on-device processing strategy.

Responsible: Team-wide collaboration

Expert Input and Validation

Psychology Center Collaboration

We ensure the clinical validity and ethical standards of our application through regular consultations with professional psychologists.

Mentor Support

We continuously develop our solution with feedback from our supervisor, Innovation expert and Global Turks AI mentors.

01

Ethical Evaluation

Expert approval on data privacy and user rights

02

User Experience

UX optimization in psychological support processes

03

Solution Validation

Clinical efficacy and technical feasibility tests

Why ScubaMind?

Continuous & Passive Monitoring

Always-on monitoring through passive behavioral signals, without interrupting daily life.

Early Awareness, Not Crisis Response

Focused on recognizing early changes in mental wellbeing before critical stages.

Minimal User Burden

No daily surveys or constant manual input required from users.

Privacy-First by Design

Sensitive behavioral data is handled securely with a strong emphasis on user privacy and ethical data use.

Why It Matters

Mental health is shaped by everyday behaviors, not only moments of crisis. ScubaMind supports early awareness by quietly and continuously monitoring behavioral patterns, enabling sustainable mental wellbeing support.

Key Differentiator

Unlike traditional mental health applications, ScubaMind emphasizes prevention through passive monitoring rather than reactive intervention.

Thank You!

We appreciate your interest in ScubaMind's innovative approach to mental wellbeing through passive insights.

Together, we can shape the future of mental health and provide sustainable well-being for everyone.

For investment and collaboration opportunities you can visit our website:

https://scubamind.netlify.app/